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GATE ECE · ENGINEERING MATHEMATICS · CHAPTER-WISE PYQS

Vector Calculus

GATE ECE Engineering Mathematics · Previous Year Questions with Answer Key

25	18	1993 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	14 questions
Marks 2	11 questions
Answer key	at the end

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Marks 1 14 questions

Q1. GATE ECE 2026

A surface is given by $z^2 = 2x^2 - y^2$ and $\vec{n}$ and $-\vec{n}$ are unit normal vectors to the surface at the point $\vec{P} = \hat{i} + \sqrt{2}\hat{k}$.

Which of the following vectors can be $\vec{n}$, where $\hat{i}, \hat{j}$ and $\hat{k}$ and are the unit vectors along x, y and z axes, respectively?

(A) $\hat{i} - \sqrt{2}\hat{k}$

(B) $\frac{2}{3}\hat{i} - \frac{1}{3}\hat{k}$

(C) $\sqrt{2}\hat{i} - \sqrt{3}\hat{k}$

(D) $\frac{\sqrt{2}\hat{i} - \hat{k}}{\sqrt{3}}$

Open solution

Q2. GATE ECE 2024

Let $\rho(x, y, z, t)$ and $u(x, y, z, t)$ represent density and velocity, respectively, at a point (x, y, z) and time t. Assume $\frac{\partial \rho}{\partial t}$ is continuous. Let V be an arbitrary volume in space enclosed by the closed surface S and $\hat{n}$ be the outward unit normal of S. Which of the following equations is/are equivalent to $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$?

(A) $$\int_V \frac{\partial \rho}{\partial t} dv = -\oint_S \rho u \cdot \hat{n}\, ds$$

(B) $$\int_V \frac{\partial \rho}{\partial t} dv = \oint_S \rho u \cdot \hat{n}\, ds$$

(C) $\int_V \frac{\partial \rho}{\partial t} dv = -\int_V \nabla \cdot (\rho u)\, dv$

(D) $\int_V \frac{\partial \rho}{\partial t} dv = \int_V \nabla \cdot (\rho u)\, dv$

Open solution

Q3. GATE ECE 2023

Let $v_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ be two vectors. The value of the coefficient α in the expression $v_1 = \alpha v_2 + e$, which minimizes the length of the error vector e, is

(A) $\frac{7}{2}$ (B) $\frac{-2}{7}$ (C) $\frac{2}{7}$ (D) $\frac{-7}{2}$

Open solution

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Q4. GATE ECE 2023

The rate of increase, of a scalar field $f(x, y, z) = xyz$, in the direction $v = (2, 1, 2)$ at a point (0,2,1) is

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) 2 (D) 4

Open solution

Q5. GATE ECE 2020

For a vector field $\vec{A}$, which one of the following is FALSE?

(A) $\vec{A}$ is solenoid if $\nabla \cdot \vec{A} = 0$.

(B) $\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$

(C) $\nabla \times \vec{A}$ is another vector field.

(D) $\vec{A}$ is irrotational if $\nabla^2 \vec{A} = 0$.

Open solution

Q6. GATE ECE 2017 Set 2

The smaller angle (in degrees) between the planes $x + y + z = 1$ and $2x - y + 2z = 0$ is _____.

Open solution

Q7. GATE ECE 2015 Set 1

A vector $\overrightarrow{P}$ is given by $\overrightarrow{P} = x^3 y \overrightarrow{a_x} - x^2 y^2 \overrightarrow{a_y} - x^2 yz \overrightarrow{a_z}$. Which one of the following statements is TRUE?

(A) $\overrightarrow{P}$ is solenoidal, but not irrotational

(B) $\overrightarrow{P}$ is irrotational, but not solenoidal

(C) $\overrightarrow{P}$ is neither solenoidal nor irrotational

(D) $\overrightarrow{P}$ is both solenoidal and irrotational

Open solution

Q8. GATE ECE 2014 Set 4

The magnitude of the gradient for the function $f(x, y, z) = x^2 + 3y^2 + z^3$ at the point $(1, 1, 1)$ is _____.

Open solution

Q9. GATE ECE 2014 Set 4

The directional derivative of $f(x, y) = \frac{xy}{\sqrt{2}}(x + y)$ at $(1, 1)$ in the direction of the unit vector at an angle of $\frac{\pi}{4}$ with $y-$axis, is given by _____.

Open solution

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Q10. GATE ECE 2014 Set 2

If $\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$ and $|\vec{r}| = r$, then div $\left(r^2 \nabla (\ln r)\right)$ = _____.

Open solution

Q11. GATE ECE 2013

Consider a vector field $\vec{A}(\vec{r})$. The closed loop line integral $\oint \vec{A} \bullet \vec{dl}$ can be expressed as

(A)

$$\oiint \left(\nabla \times \vec{A}\right) \bullet \vec{ds}$$

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over the closed surface bounded by the loop

(B)

$$\oiiint \left(\nabla \bullet \vec{A}\right) dv$$

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over the closed volume bounded by the loop

(C)

$$\int\int\int \left(\nabla \bullet \vec{A}\right) dv$$

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over the open volume bounded by the loop

(D)

$$\int\int \left(\nabla \times \vec{A}\right) \bullet \vec{ds}$$

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over the open surface bounded by the loop

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Q12. GATE ECE 2011

Open solution

Consider a closed surface $'S'$ surrounding a volume V. If $\vec{r}$ is the position vector of a point inside S with $\vec{n}$ the unit normal on $'S'$, the value of the integral

$$\oiint 5\vec{r}.\hat{n}ds$$

ExamSide.Com

is

(A) $3V$ (B) $5V$ (C) 10 (D) $15V$

Q13. GATE ECE 2005

Open solution

$\nabla \times (\nabla \times P)$ where P is a vector is equal to

(A) $P \times \nabla \times P \to \nabla^2 P$ (B) $\nabla^2 P + \nabla(\nabla . P)$

(C) $\nabla^2 P + (\nabla \times P)$ (D) $\nabla(\nabla . P) \to \nabla^2 P$

Q14. GATE ECE 1993

Open solution

If the linear velocity $\vec{V}$ is given by $\vec{V} = x^2 y\vec{i} + xyz\vec{j} - yz^2\vec{k}$ then the angular velocity $\vec{W}$ at the point $(1, 1, -1)$ is _____.

Marks 2 11 questions

Q15. GATE ECE 2024

Open solution

Let F_1, F_2, and F_3 be functions of (x, y, z). Suppose that for every given pair of points A and B in space, the line integral $\int_C (F_1 dx + F_2 dy + F_3 dz)$ evaluates to the same value along any path C that starts at A and ends at B. Then which of the following is/are true?

(A) For every closed path Γ, we have $\oint_\Gamma (F_1 dx + F_2 dy + F_3 dz) = 0$.

(B) There exists a differentiable scalar function $f(x, y, z)$ such that $F_1 = \frac{\partial f}{\partial x}, F_2 = \frac{\partial f}{\partial y}, F_3 = \frac{\partial f}{\partial z}$.

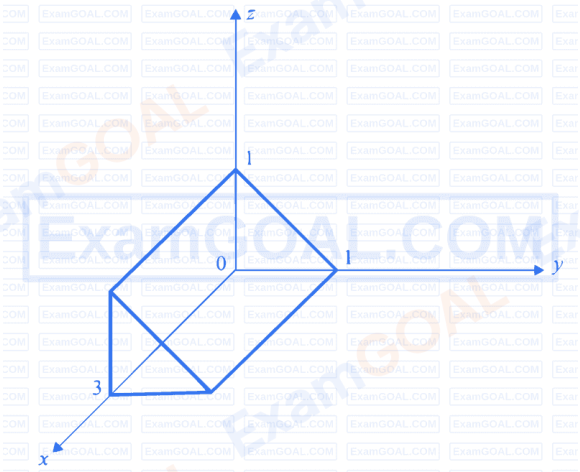
(C) $\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 0$.

(D) $\frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$.

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Q16. GATE ECE 2020

For the solid S shown below, the value of $\iiint_S x dx dy dz$ (rounded off to two decimal places) is ____ .



Open solution

Q17. GATE ECE 2017 Set 1

If the vector function

$\overrightarrow{F} = \hat{a}_x (3y - k_1 z) + \hat{a}_y (k_2 x - 2z) - \hat{a}_z (k_3 y + z)$

is irrotational, then the values of the constants k_1, k_2 and k_3 respectively, are

(A) $0.3, -2.5, 0.5$ (B) $0.0, 3.0, 2.0$ (C) $0.3, 0.33, 0.5$ (D) $4.0, 3.0, 2.0$

Open solution

Q18. GATE ECE 2017 Set 1

Let $\mathrm{I} = \int_c (2z\,dx + 2y\,dy + 2x\,dz)$ where x, y, z are real, and let C be the straight line segment from point $A : (0, 2, 1)$ to point $B : (4, 1, -1)$. The value of I is _____.

Open solution

Q19. GATE ECE 2016 Set 2

Suppose C is the closed curve defined as the circle $x^2 + y^2 = 1$ with C oriented anti-clockwise. The value of $\oint (xy^2\,dx + x^2 y\,dy)$ over the curve C equals _____.

Open solution

Q20. GATE ECE 2014 Set 4

Given $\overrightarrow{F} = z\hat{a}_x + x\hat{a}_y + y\hat{a}_z$. If S represents the portion of the sphere $x^2 + y^2 + z^2 = 1$ for $z \geq 0$, then $\int_s \left(\nabla \times \overrightarrow{F}.\right) \overrightarrow{ds}$ is _____.

Open solution

Q21. GATE ECE 2013

The divergence of the vector field $\overrightarrow{A} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$ is

(A) 0 (B) $1/3$ (C) 1 (D) 3

Open solution

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Q22. GATE ECE 2012

Open solution

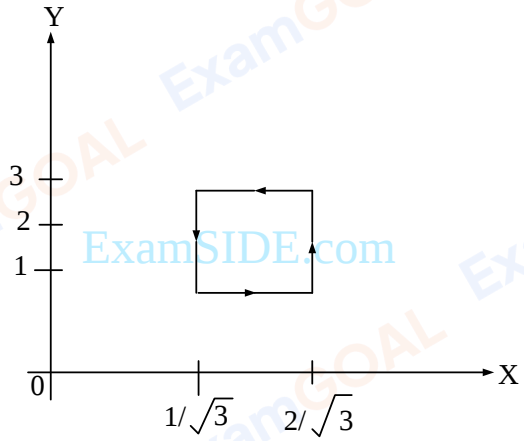
The direction of vector A is radially outward from the origin, with $|A| = K r^n$ where $r^2 = x^2 + y^2 + z^2$ and K is constant. The value of n for which $\nabla . A = 0$ is

(A) -2 (B) 2 (C) 1 (D) 0

Q23. GATE ECE 2010

Open solution

If $\overrightarrow{A} = xy\,\hat{a}_x + x^2\hat{a}_y$ then $\oint \overrightarrow{A}.\overrightarrow{dr}$ over the path shown in the figure is



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(A) 0 (B) $\frac{2}{\sqrt{3}}$ (C) 1 (D) $2\sqrt{3}$

Q24. GATE ECE 2009

Open solution

If a vector field $\overrightarrow{V}$ is related to another field $\overrightarrow{A}$ through $\overrightarrow{V} = \nabla \times \overrightarrow{A}$, which of the following is true?
Note: C and S_C refer to any closed contour and any surface whose boundary is C.

(A) $\oint_C \overrightarrow{V}.\overrightarrow{dl} = \int\int_{S_C} \overrightarrow{A}.\overrightarrow{ds}$

(B) $\oint_C \overrightarrow{A}.\overrightarrow{dl} = \int_{S_C}\int \overrightarrow{V}.\overrightarrow{ds}$

(C) $\oint_C \nabla \times \vec{V}.\overrightarrow{dl} = \int_{S_C}\int \nabla \times \vec{A}.\overrightarrow{ds}$

(D) $\oint_C \nabla \times \vec{A}.\overrightarrow{dl} = \int\int_{S_C} \vec{V}.\overrightarrow{ds}$

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Q25. GATE ECE 2008

Open solution

Consider points P and Q in $xy-$plane with $P = (1, 0)$ and $Q = (0, 1)$. The line integral $2\int_{P}^{Q}(x\,dx + y\,dy)$ along the semicircle with the line segment PQ as its diameter

(A) is -1

(B) is 0

(C) 1

(D) depends on the direction (clockwise (or) anti-clockwise) of the semi circle

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Answer Key

Marks 1

1. **D**	2. **A, C**	3. **C**	4. **B**	5. **D**
6. **54 to 56**	7. **A**	8. **7**	9. **3**	10. **3**
11. **D**	12. **D**	13. **D**	14. $-\left(\hat{i}+\hat{k}\right)$	

Marks 2

15. **A, B, D**	16. **2.25**	17. **B**	18. **-11**	19. **0**
20. **3.14**	21. **D**	22. **A**	23. **C**	24. **B**
25. **B**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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